JATIN SATYAM

MS Robotics & Autonomous Systems @ ASU · Mechanical Design · Thermal & Structural Analysis

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EDUCATION

M.S. Robotics & Autonomous Systems (Systems Eng.), Arizona State University · GPA 3.67/4.0 — Expected May 2027

B.E. Mechanical Engineering with Major (Honors) in Electric Vehicles, University of Mumbai — 2020 – 2024

EXPERIENCE

Research Assistant — Aug 2026 – Present

Robotics Systems Instructional Lab, Arizona State University

- Built and commissioned a second bench of a two-robot semiconductor wafer-handling cell (SCARA and six-axis arms, conveyor, Micro850 PLC), and turned both benches into labs for an ASU short-course manual
- Used the cell's NVIDIA Isaac Sim digital twin to find 2 interlock changes that cut the real cell's cycle time 26% (139.9 to 103.9 s, reproduced in 3 of 3 runs); its wafer contact physics matched the bench (2.6 mm slide predicted, 2.1-2.5 mm measured)

Mechatronics Engineer — Jan 2025 – Jul 2025

Paras Defence & Space Technologies Ltd., Navi Mumbai, India · full-time at graduate-engineer-trainee scope

- Designed and iterated Gifford-McMahon cryocooler mechanisms in SolidWorks and NX, refining moving-assembly tolerances and interfaces for reliable cold-head displacer motion and manufacturability
- Led performance testing of 3 cryogenic cooling assemblies across 25+ instrumented thermal runs, turning loss data into mechanism-level design changes that cut prototype rework 12%
- Formulated 8 test plans for electromechanical assembly functionality, cutting deployment errors 15% with consistent pass/fail criteria
- Built a vacuum-insulated LN2 generation unit sized to minimize parasitic heat leak, and ran a UAV structural lightweighting study that cut mass 8% through material and topology changes
- Delivered 30+ CAD models and drawing revisions and managed 3 project lifecycles in PLM, cutting design-review turnaround 20% with no version drift between design and fabrication

CFD Engineer — Aug 2024 – Nov 2024

MARS Exploration Pvt. Ltd (remote) · AS9100D-certified aerospace engineering firm

- Performed 3D CFD of NACA 23012 vs 4412 airfoils in ANSYS Fluent 2024 R2 (10-degree AoA, 48 m/s), from boundary-layer meshing to vortex post-processing; NACA 4412 delivered higher lift

R&D Engineer — May 2023 – Aug 2023

TriQual Cryogenics Pvt Ltd, Navi Mumbai, India

- Contributed to mechanical design and vibration control for Stirling-cycle cryocoolers near 70 K and gyro motor and helium compressor development (Make-in-India); delivered every task in a short span, speeding up the team's R&D

PROJECTS

Gyro Motor Rotor Development ↗ · *SolidWorks · ANSYS Rotordynamics · PHOTON+ FFT* — Industry project · 2024

- Designed a gyro motor rotor for defense stabilized platforms: 8 iterations (straight and inclined bar, 8-11 mm, EN24/EN8) screened in ANSYS for critical speeds, damped frequencies and stress at 28,000 rpm / 466.67 Hz
- Chose the 9 mm straight-bar design of the 8 on a 5,803 rpm critical-speed margin; built it and ran it at its full 28,000 rpm on an instrumented bench, with FFT placing the critical speed safely below at 19,462 rpm

SipSafe Smart Cup - Assistive Device for Dysphagia Patients · *Mechanical Design · Prototyping* — ASU · 2026

- Owned the concept and complete mechanical design of a dysphagia smart cup, built by a 5-person team with Dignity Health

SKILLS

CAD & Mechanical Design: SolidWorks, Siemens NX, Fusion 360, AutoCAD, GD&T, DFM, Tolerance Analysis

Simulation & Analysis: ANSYS (Fluent, Modal/Rotordynamics), FEA, CFD, Thermal & Structural Analysis, Isaac Sim (PhysX contact)

Test & Instrumentation: Test Rig Design, DAQ, LabVIEW, FFT Signal Analysis, Accelerometry, Design of Experiments

Manufacturing, Systems & Tools: PLM & Configuration Management, FMEA, Root-Cause Analysis, V&V, Prototype Development, PLC Commissioning (Micro850), MATLAB/Simulink, Python, ROS 2 Jazzy, Gazebo Harmonic

AWARDS & RECOGNITION

- 2nd Prize, National Level Project Poster Competition, IEI-FCRIT 2024 - gyro motor rotor (CALIBRE 2k24)
- Co-inventor, invention disclosure filed with ASU's Skysong Innovations - SipSafe Smart Cup, 2026
- Co-author (team lead): Development of Solar-Powered Electric Vehicle Prototype, CPIE 2023 proceedings, Springer, 2024